STATISTICS

SOME PROOFS FOR REGRESSION ANALYSIS

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Regression Line

Population Parameter	Sample statistic
β_0	b_0
β_1	b_1
$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$	$\hat{Y}_i = b_0 + b_1 X_i$

Here;

- Y_i: the observation point the dependent variable and it is random variable
- X_i: the observation point the independent variable and it is not random variable
- β_0: the parameter of the population and it is not random
- β_1: the parameter of the population and it is not random
- b_0: the statistic of the sample and it is random
- b_1: the statistic of the sample and it is random
- ε_i: error term and it is random. Here, we assume that ε_i's are independent and

$$\varepsilon_i \sim N(0, \sigma^2)$$

Proofs for Regression Analysis

Statistics

Normal equations of the line are:

$$b_1 = \frac{\sum_{i=1}^{n} X_i Y_i - \frac{(\sum_{i=1}^{n} X_i)(\sum_{i=1}^{n} Y_i)}{n}}{\sum_{i=1}^{n} {X_i}^2 - \frac{\left(\sum_{i=1}^{n} X_i\right)^2}{n}} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{n}(X_i - \bar{X})^2}$$

$$b_0 = \bar{Y} - b_1 \bar{X}$$

Theorem 1: $E[b_1] = \beta_1$ (b_1 is an unbiased estimator of β_1)

Proof 1:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \qquad \sum_{i=1}^{n}(X_i - \bar{X}) = 0$$

$$b_1 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})\overbrace{(Y_i} - \bar{Y})}{\sum_{i=1}^{n}(X_i - \bar{X})^2} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(\beta_0 + \beta_1 X_i + \varepsilon_i - \bar{Y})}{\sum_{i=1}^{n}(X_i - \bar{X})^2} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(\beta_1 X_i + \varepsilon_i)}{\sum_{i=1}^{n}(X_i - \bar{X})^2}$$

$$\sum_{i=1}^{n}(X_i - \bar{X})X_i = \sum_{i=1}^{n}(X_i - \bar{X})^2 \quad \Rightarrow \quad b_1 = \beta_1 + \frac{\sum_{i=1}^{n}((X_i - \bar{X})\varepsilon_i)}{\sum_{i=1}^{n}(X_i - \bar{X})^2} \qquad E[\varepsilon_i] = 0$$

$$\Rightarrow \quad E[b_1] = E\left[\beta_1 + \frac{\sum_{i=1}^{n}((X_i - \bar{X})\varepsilon_i)}{\sum_{i=1}^{n}(X_i - \bar{X})^2}\right] = E[\beta_1] + \frac{\sum_{i=1}^{n}[(X_i - \bar{X})\overbrace{E[\varepsilon_i]}]}{\sum_{i=1}^{n}(X_i - \bar{X})^2} = E[\beta_1] = \beta_1$$

Proofs for Regression Analysis

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Theorem 2: $E[b_0] = \beta_0$ (b_0 is an unbiased estimator of β_0)

Proof 2:

$$b_0 = \bar{Y} - b_1\bar{X}$$

$$\Rightarrow \quad E[b_0] = E[\bar{Y} - b_1\bar{X}] = E[\bar{Y}] - \bar{X}\overbrace{E[b_1]}^{E[b_1] = \beta_1} = \underbrace{\beta_0 + \beta_1\bar{X}} - \bar{X}\beta_1 = \beta_0$$

$$E[\bar{Y}] = E\left[\frac{1}{n}\sum_{i=1}^{n} Y_i\right] = \frac{1}{n}\sum_{i=1}^{n} E[Y_i] = \frac{1}{n}\sum_{i=1}^{n} E[\beta_0 + \beta_1 X_i + \varepsilon_i] = \beta_0 + \beta_1\bar{X}$$

$$E[\varepsilon_i] = 0$$

Proofs for Regression Analysis

Statistics

Theorem 3: $Var[b_1] = \frac{\sigma^2}{\sum_{i=1}^{n}(X_i - \bar{X})^2} = \frac{\sigma^2}{\sum_{i=1}^{n} X_i^2 - \frac{(\sum_{i=1}^{n} X_i)^2}{n}}$

Proof 3:

In proof 1, we obtained that $b_1 = \beta_1 + \frac{\sum_{i=1}^{n}(X_i - \bar{X})\varepsilon_i}{\sum_{i=1}^{n}(X_i - \bar{X})^2}$

$$\Rightarrow \quad Var[b_1] = Var\left[\beta_1 + \frac{\sum_{i=1}^{n}(X_i - \bar{X})\varepsilon_i}{\sum_{i=1}^{n}(X_i - \bar{X})^2}\right]$$

ε_i's are independent

$$= \underbrace{Var[\beta_1]}_{0} + \frac{Var[\sum_{i=1}^{n}(X_i - \bar{X})\varepsilon_i]}{\left(\sum_{i=1}^{n}(X_i - \bar{X})^2\right)^2}$$

$$\sigma^2 = Var[Y_i] = Var[\beta_0 + \beta_1 X_i + \varepsilon_i] = Var[\varepsilon_i]$$

$Var[\varepsilon_i] = \sigma^2$

$$= \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2 Var(\varepsilon_i)}{\left(\sum_{i=1}^{n}(X_i - \bar{X})^2\right)^2} = \frac{\sigma^2}{\sum_{i=1}^{n}(X_i - \bar{X})^2}$$

Proofs for Regression Analysis

Statistics

Theorem 4: $Var[b_0] = \left(\frac{1}{n} + \frac{\bar{X}^2}{\sum_{i=1}^{n}(X_i - \bar{X})^2}\right)\sigma^2 = \frac{\sum_{i=1}^{n} X_i^{\,2}}{n\sum_{i=1}^{n} X_i^{\,2} - \left(\sum_{i=1}^{n} X_i\right)^2}\sigma^2$

Proof 4:

$$b_0 = \bar{Y} - b_1\bar{X}$$

$$\Rightarrow \quad Var[b_0] = Var[\bar{Y} - b_1\bar{X}] = Var[\bar{Y}] + (-1)^2\bar{X}^2 Var[b_1] = \frac{\sigma^2}{n} + \frac{\bar{X}^2\sigma^2}{\sum_{i=1}^{n}(X_i - \bar{X})^2}$$

$$Var[b_1] = \frac{\sigma^2}{\sum_{i=1}^{n}(X_i - \bar{X})^2}$$

$$Var[\bar{Y}] = Var\left[\frac{1}{n}\sum_{i=1}^{n} Y_i\right] = \frac{1}{n^2}\sum_{i=1}^{n} Var[Y_i] = \frac{1}{n^2}n\sigma^2 = \frac{\sigma^2}{n}$$

$$Var[Y_i] = Var[\beta_0 + \beta_1 X_i + \varepsilon_i] = Var[\varepsilon_i] = \sigma^2$$